



Workspace camera
Puzzle solution
Observed task progress

Task context

1. Physiological processing



Wrist sensor
HR via Bluetooth

Signal quality
Valid HR and sensor contact
Sufficient usable samples

Current window
Latest 15 s
Median HR

Reference
Preceding 60 s
Median HR
or calibrated
baseline

Heart-rate rise
 $\Delta HR = HR_{cur} - HR_{ref}$

Arousal flag
Raised when $\Delta HR \geq 5$ bpm

2. Assistance timing

Assigned condition

Constant
Assistance every 30 s

Adaptive
Assistance whenever
arousal is flagged

No assistance
Hints and robot
actions disabled

3. Wizard-of-Oz delivery

Researcher dashboard
Select a hint and/or robot
action using task context

Text hint
Piece location, ori-
entation or adjacency

Robot operator
Executes the selected
piece-specific program

4. Participant interaction

Participant
Solves the tan-
gram puzzle

Participant laptop
For hints, timer
and questionnaires

Robotic arm
Retrieves the
selected piece

Piece at the drop spot

Questionnaire responses

Local study server

Shared session state, interface synchronization and recording

Study records

Task outcomes, physiological data, intervention events and questionnaire responses

→ Signal processing and assistance

- - -> Task context and responses